

Data Analysis With Pandas

Submitted By:
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Objective:
To analyze student data using the Pandas library in Python.

Tools Used:

- Python
- Replit
- Pandas Library

Project Description:
This project demonstrates student data analysis using the Pandas library in Python. The program creates student records in table format, performs calculations such as average marks, highest marks, lowest marks, and displays student information.

Python Code:

```
import pandas as pd

data = {
    "Name": ["Manasa", "Anagha", "Harsha"],
    "Marks": [95, 88, 76],
    "City": ["Pileru", "Hyderabad", "Tirupati"]
}

df = pd.DataFrame(data)

print("===== Student Data Analysis =====\n")

print("Student Records:\n")
print(df)

print("\nTotal Students:")
print(len(df))

print("\nAverage Marks:")
```

```
print(df["Marks"].mean())

print("\nHighest Marks:")
print(df["Marks"].max())

print("\nLowest Marks:")
print(df["Marks"].min())

print("\nStudent with Highest Marks:")
print(df.loc[df["Marks"].idxmax()])

print("\nStudent with Lowest Marks:")
print(df.loc[df["Marks"].idxmin()])
```

Output:

===== Student Data Analysis =====

Student Records:

	Name	Marks	City
0	Manasa	95	Pileru
1	Anagha	88	Hyderabad
2	Harsha	76	Tirupati

Total Students:

3

Average Marks:

86.33333333333333

Highest Marks:

95

Lowest Marks:

76

Features Implemented:

- Data analysis using Pandas
- DataFrame creation

- Average marks calculation
- Highest and lowest marks analysis
- Student data management

Conclusion:

Successfully implemented student data analysis using the Pandas library in Python.

```
Data Insights + Upgrade Preview x main.py x Shell x +
main.py > ...
1 import pandas as pd
2
3 data = {
4     "Name": ["Manasa", "Anagha", "Harsha"],
5     "Marks": [95, 88, 76],
6     "City": ["Pileru", "Hyderabad", "Tirupati"]
7 }
8
9 df = pd.DataFrame(data)
10
11 print("==== Student Data Analysis =====")
12
13 print("Student Records:\n")
14 print(df)
15
16 print("\nTotal Students:")
17 print(len(df))
18
19 print("\nAverage Marks:")
20 print(df["Marks"].mean())
21
22 print("\nHighest Marks:")
23 print(df["Marks"].max())
24
25 print("\nLowest Marks:")
26 print(df["Marks"].min())
27
28 print("\nStudent with Highest Marks:")
29 print(df.loc[df["Marks"].idxmax()])
30
31 print("\nStudent with Lowest Marks:")
32 print(df.loc[df["Marks"].idxmin()])
```

```
Data Insights + Upgrade Preview x main.py x Shell x +
~/workspace:python main.py
==== Student Data Analysis =====
Student Records:
   Name  Marks  City
0  Manasa    95  Pileru
1  Anagha    88  Hyderabad
2  Harsha    76  Tirupati

Total Students:
3

Average Marks:
86.33333333333333

Highest Marks:
95

Lowest Marks:
76

Student with Highest Marks:
Name    Manasa
Marks    95
City    Pileru
```